IN PROGRESS DOCUMENT AI & GENAI

ClearCustoms AI

Automated parsing and semantic extraction for global trade documentation.

Active Development Phase

THE PROBLEM

Trade Friction

- **Unstructured Chaos:** Thousands of varied invoice formats globally.
- **Manual Bottlenecks:** Slow, error-prone human data entry.
- **Compliance Risks:** Mistyped HS codes trigger costly customs delays.



Core Objectives



Layout Independence

Extract fields without rigid templates.



Semantic Reasoning

Understand context via Generative LLMs.



Structured Output

Generate clean JSON for ERP integration.

Planned Technology Stack



Python



Tesseract / EasyOCR



LayoutLM



LLM Integration

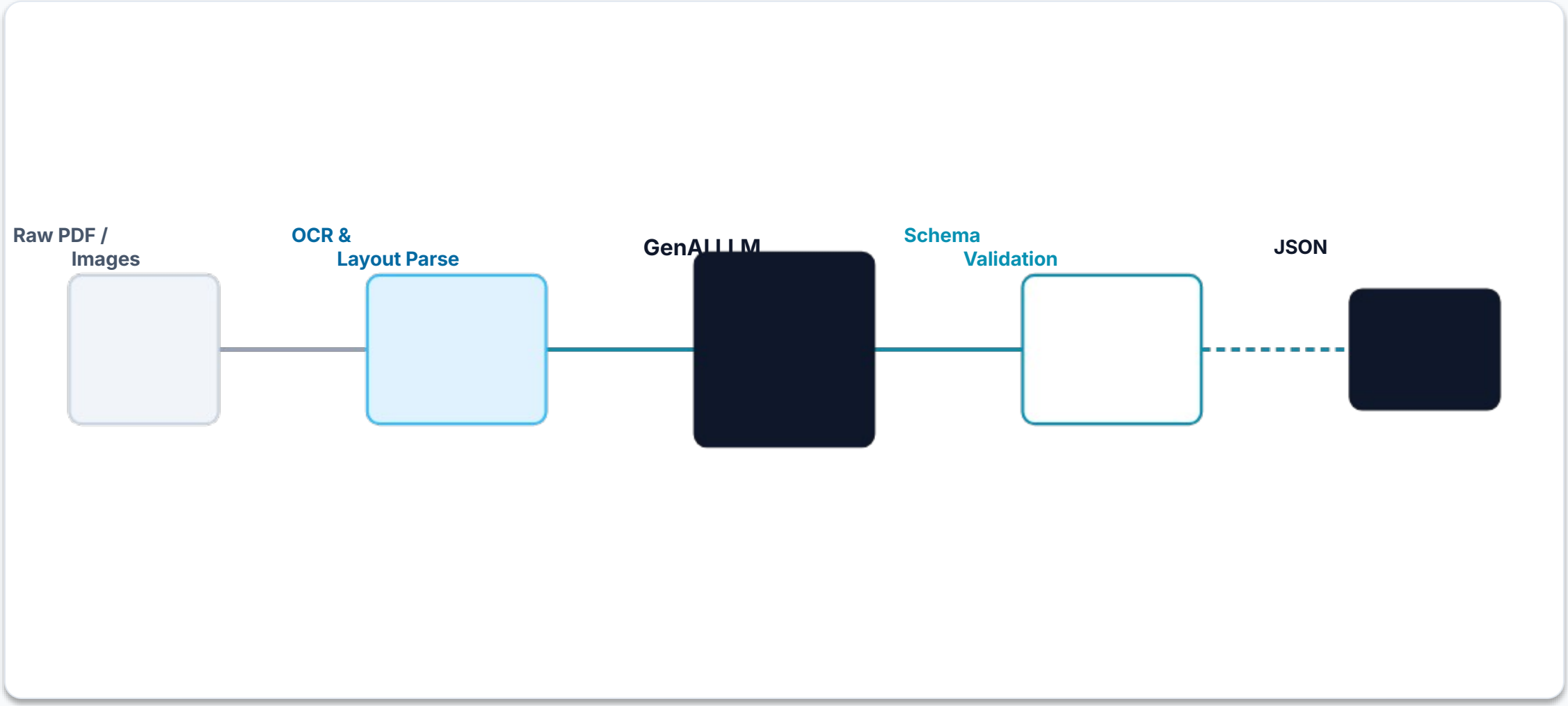


FastAPI



React UI

Extraction Pipeline Architecture



Clearance Workflow



1. Ingest

Receive scanned document.



2. Process

Deskew and OCR bounding boxes.



3. Extract

LLM entity recognition.



4. Validate

Apply customs rules.

Dataset & Preprocessing

Target Documents

- Commercial Invoices.
- Bills of Lading (BoL).
- Detailed Packing Lists.

Vision Preprocessing

- Automated deskewing algorithms.
- Adaptive binarization.
- Background noise reduction.

Hybrid AI Pipeline

Vision & Layout

Goal: Spatial awareness.

Maps text coordinates to preserve the visual hierarchy of complex nested tables.

Generative Parsing

Goal: Semantic extraction.

LLM interprets key-value pairs regardless of visual placement or varying terminology.



Review Dashboard

Human-in-the-loop interface comparing source documents alongside structured AI extractions.

- **Confidence Scoring.**
- **One-click correction.**
- **Direct ERP export.**

Current Challenges

OCR Limitations

- Low DPI Scans: Illegible micro-text.
- Watermarks: Interference with character recognition.

Generative Parsing

- Hallucinations: Inventing missing HS codes.
- Nested Tables: Misalignment of multi-page line items.

Technical Q&A

1. Why not use standard Regex rules?

Invoices have infinite layout variations; Regex is too brittle.

2. How to prevent LLM hallucinations on numbers?

Enforce strict JSON schemas and ground extraction via temperature=0.

3. Advantage of LayoutLM over standard OCR?

Embeds spatial coordinates alongside text for 2D visual context.

4. Handling multi-page tables?

Chunking by visual separators and merging entity arrays post-extraction.

5. How do you validate the output?

Cross-referencing extracted totals against mathematical sums of line items.

Frictionless Global Trade.

Transforming legacy supply chain documents into structured, actionable intelligence.



Repository: Private / WIP

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Image Sources



https://d3lkc3n5th01x7.cloudfront.net/wp-content/uploads/2023/12/28004739/AI-in-Customs-Clearance.png

Source: www.leewayhertz.com



https://media.licdn.com/dms/image/D4D12AQExyO-o8F-PMA/article-cover_image-shrink_720_1280/0/1691157956041?e=2147483647&v=beta&t=37yT-F3o3g4oF6H8iR-uH-j1V4nE9V_D_3M190yA8zE

Source: www.linkedin.com



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